

Databases -- Guided Notes ANSWER KEY

Unit tp-4.5 | CompTIA Network+ N10-009 Obj. FC0-U71 4.5 | N10-008 FC0-U71 4.5

For instructor use / distribute after students complete the notes.

Question 1

A database is an organized collection of _____ designed for efficient retrieval and manipulation. A _____ (DBMS) is software that manages the database -- examples include MySQL, PostgreSQL, and Microsoft SQL Server. Users interact with the DBMS using _____ (Structured Query Language).

Answer: Data; DBMS (Database Management System); SQL.

Question 2

In a relational database, data is stored in _____ (rows and columns). Each row is called a _____ and each column is called a _____. A _____ key is a column whose value uniquely identifies each row. A _____ key in one table references the primary key of another table.

Answer: Tables; record (row); field (column/attribute); primary; foreign.

Question 3

The four basic database operations are described by the acronym _____: _____ (add new records), _____ (retrieve data), _____ (modify existing data), and _____ (remove records).

Answer: CRUD; Create; Read; Update; Delete.

Question 4

A SELECT query retrieves data from a database. Write the basic SQL syntax to retrieve all records from a table named students:

_____ * _____ students;

To retrieve only students with grade = 'A', add a _____ clause:

_____ * FROM students _____ grade = 'A';

*Answer: SELECT * FROM students; WHERE; SELECT * FROM students WHERE grade = 'A';*

Question 5

Database normalization reduces _____ (storing the same data in multiple places) by organizing tables so that each piece of data is stored _____ once. First normal form (1NF) requires that each column holds _____ values. Normalization improves _____ consistency and reduces the chance of update anomalies.

Answer: Redundancy (duplication); only; atomic (single, indivisible); data.

Question 6

A NoSQL database stores data in formats other than tables, such as _____ (MongoDB), _____ (Redis), _____ (Apache Cassandra), or _____ (Neo4j). NoSQL databases are often preferred when data is _____ -structured, requires horizontal scaling, or schema changes frequently.

Answer: Documents (JSON); key-value pairs; wide-column; graphs; unstructured (semi).

Question 7

SQL Injection is a security attack where an attacker inserts malicious _____ into an input field that is then executed by the database. For example, entering ' OR '1'='1 into a login form might bypass authentication. The primary defense is using _____ statements (also called parameterized queries) which treat user input as _____ rather than executable code.

Answer: SQL code; prepared; data.

Question 8 -- Real World Application

REAL WORLD: A student builds a web application where users can search for products by name. The search box directly inserts the user's input into a SQL query like: `SELECT * FROM products WHERE name = '[input]'`. Explain the SQL injection vulnerability, show an example attack, and describe the fix.

Answer: Vulnerability: the user input is concatenated directly into the SQL string without sanitization. The database executes whatever text is embedded in the query. Attack example: if the attacker enters '; DROP TABLE products;--' in the search box, the resulting SQL becomes: `SELECT * FROM products WHERE name = "; DROP TABLE products;--'`. The semicolon ends the `SELECT` and the `DROP TABLE` runs as a second query, deleting all product data. Fix: use parameterized queries (prepared statements). The query becomes: `SELECT * FROM products WHERE name = ?` and the input value is passed separately as a parameter. The database driver treats the input as a data value -- never as executable SQL code -- so even if the attacker types `DROP TABLE`, it is treated as a literal search string, not a command.